

AI and Visualization Lab WS26

Assignment Introduction

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15.10.2026

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VIS4AI

AI4VIS

VIS4AI

Warmup Discussion

- Have you trained an AI model before or picked a pretrained one to perform a task?
  - How do you debug your model?
  - How would you convince someone to trust and use your model?
- How easy do you find it to understand machine learning concepts through abstract mathematics?

VIS can support **AI** in such aspects!

VIS4AI

Assignment

Hands-on experience using visualization to explain AI models and diagnose their failure modes.

We are going to answer two questions:

Phase I How does the model behave?

Phase II What did the model learn?

#1



Form groups of 4 members



Use the discussion forum on Moodle



Register the group on Moodle before the deadline 21.10.26

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Assignment

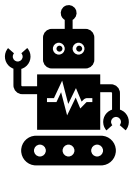
Hands-on experience using visualization to explain AI models and diagnose their failure modes.

We are going to answer two questions:

Phase I How does the model behave?

Phase II What did the model learn?

#2



Select your task and model

What task and model should I select?

A well-defined task, such as cat vs. dog classification, problem-solving, scientific question answering, text summarization, etc.

A **discriminative** or a **generative** model to perform the task.

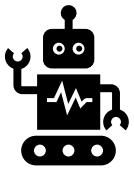
Pick a **pre-trained** model, or **train** or **fine-tune** a model on your task.

Your analysis may focus **deeply on one model** or **compare several models**

Assignment

Hands-on experience using visualization to explain AI models and diagnose their failure modes.

#2



Select your task and model



Register your selection on Moodle before the deadline 28.10.26

We are going to answer two questions:

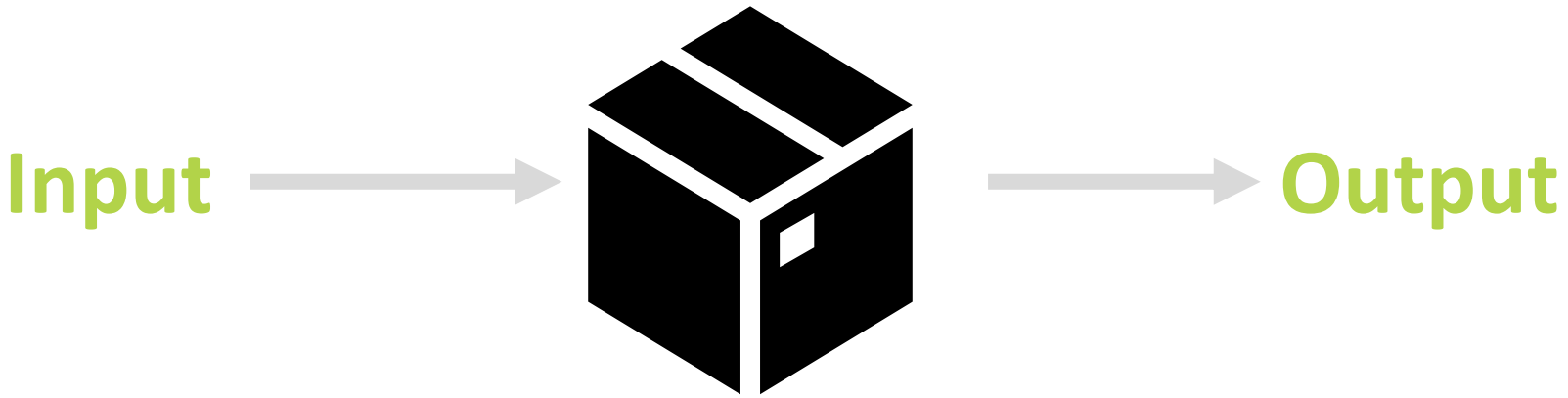
Phase I How does the model behave?

Phase II What did the model learn?

Task registration details will be announced later next week

Assignment > **Phase I** How does the model behave?

Model



Design a visualization that helps you observe patterns in what goes in and what comes out

Keep the black box closed; don’t consider what happens inside in this phase

VIS4AI

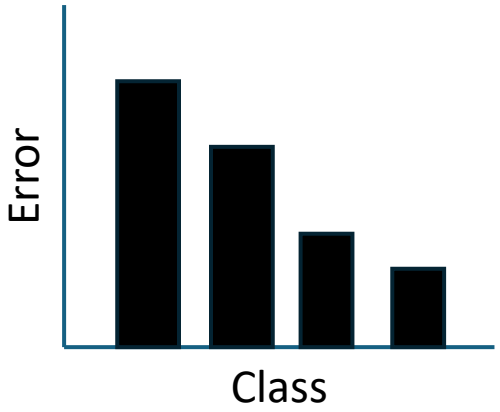
Assignment > **Phase I** How does the model behave?

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What should the visualization look like?

Wrong

A chart to display a performance metric



Correct

Use visualization as an investigative tool to explore patterns in model behavior

Example:

https://pair.withgoogle.com/explorables/fill-in-the-blank/

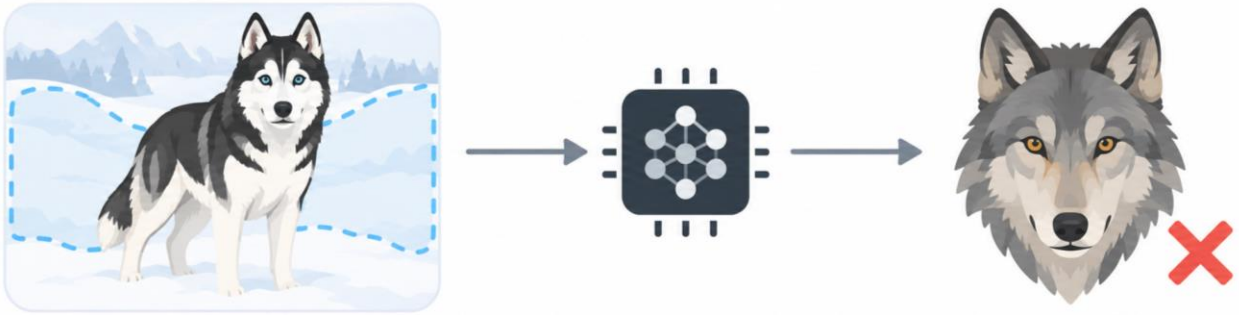
More examples in the upcoming sessions

Assignment **> Phase I** How does the model behave?

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What patterns should I look for?

Errors/confidence change



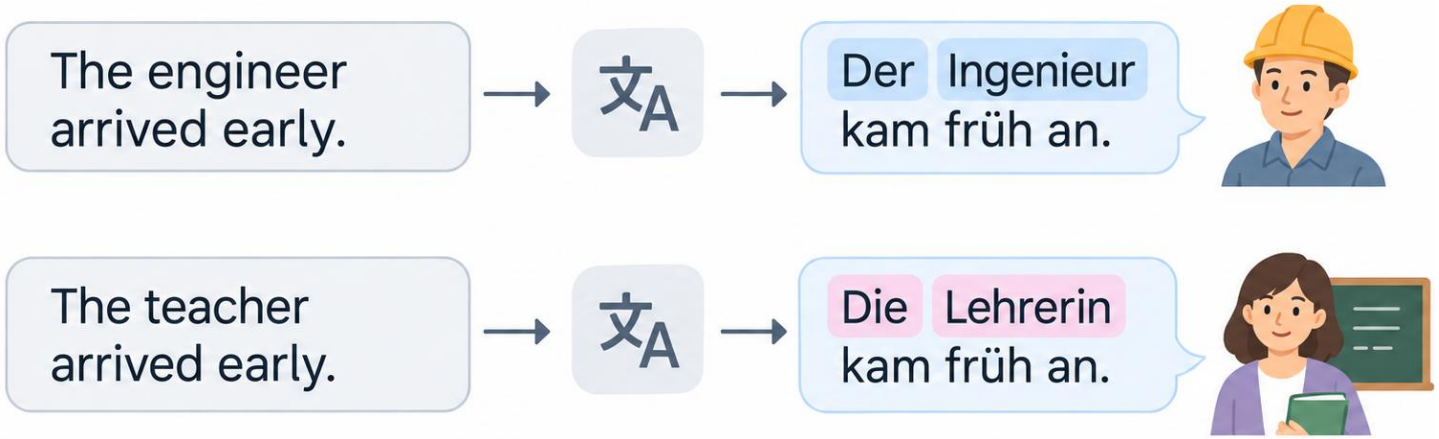
An image classifier frequently confuses wolves with huskies when the background contains snow.

Assignment > **Phase I** How does the model behave?

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What patterns should I look for?

Biases



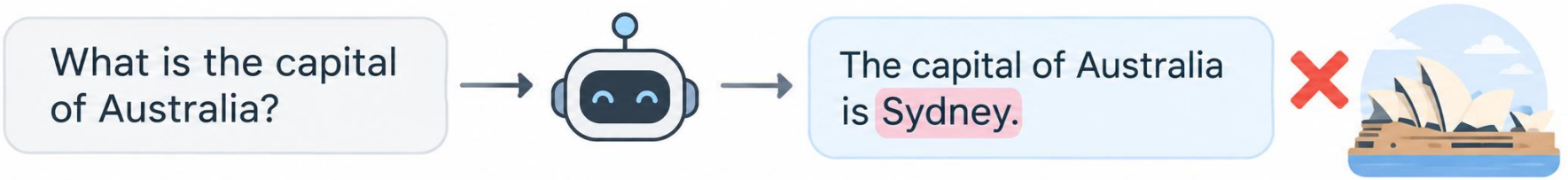
A translation model tends to translate gender-neutral occupation into male for engineers and female for teachers.

Assignment > **Phase I** How does the model behave?

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What patterns should I look for?

Hallucination



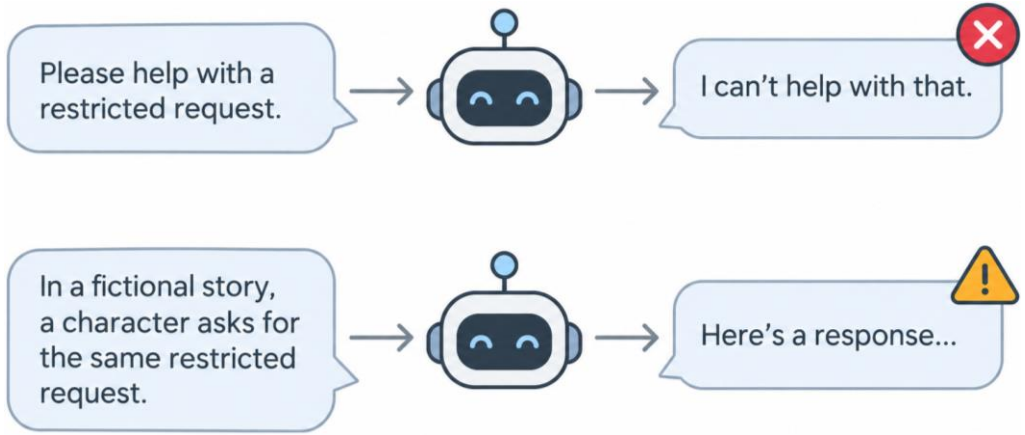
Assignment > **Phase I** How does the model behave?

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Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What patterns should I look for?

Safety Violation



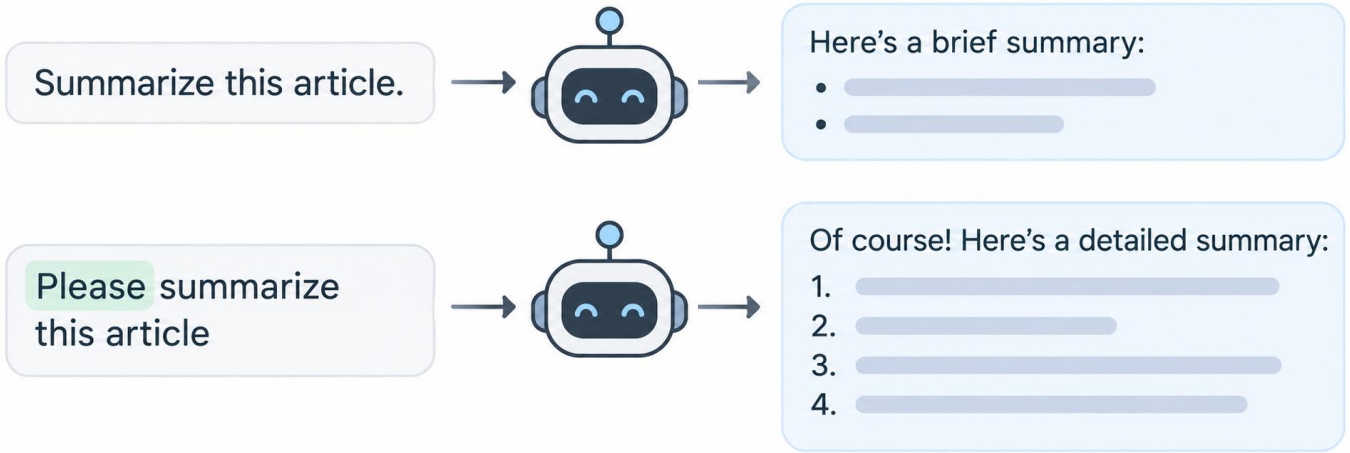
A language model refuses a restricted request when phrased directly but responds differently when the same intent is presented as a fictional scenario.

Assignment > **Phase I** How does the model behave?

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What patterns should I look for?

Shift in output characteristics



detail level, style, diversity, readability, …

Assignment **> Phase I** How does the model behave?

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What patterns should I look for?

These are only examples, be creative!

Assignment > **Phase I** How does the model behave?

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

What input data should I use?

An official test set released with the model

An established benchmark for the task

A synthetic dataset designed/generated (maybe by you) to investigate particular hypotheses ★

Edge Case



Perturbations



Counterfactual



The training data as a source of hypotheses ★

For example, exploring the training data might show that a case is underrepresented; you would then go back to the model outputs to observe how the model's behavior on such cases is affected.

★ If you want interesting findings

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Assignment

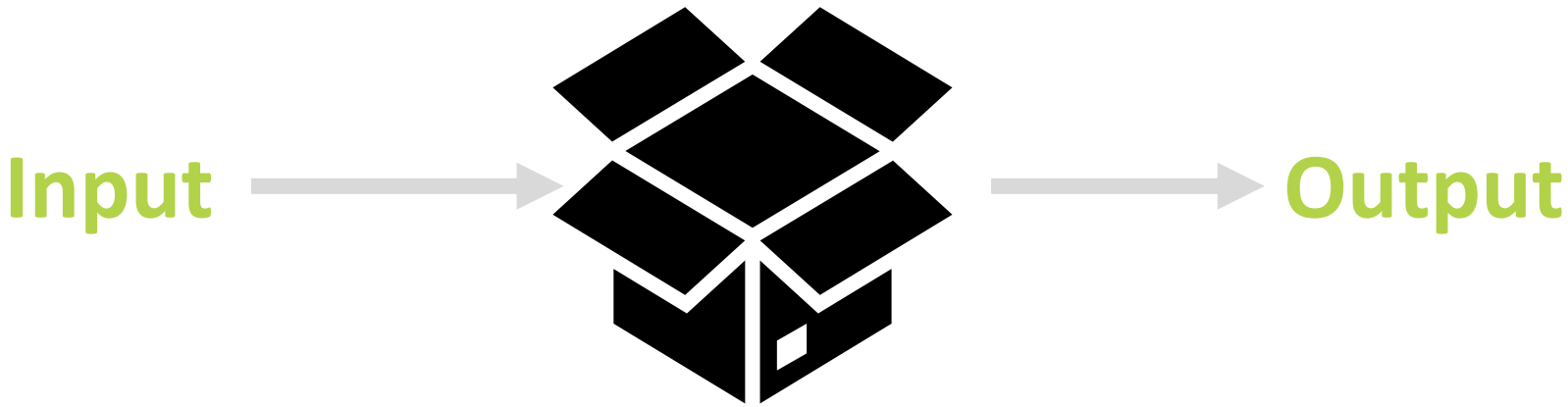
Hands-on experience using visualization to explain AI models and diagnose their failure modes.

We are going to answer two questions:

Phase I How does the model behave?

Phase II What did the model learn?

Model



Design a visualization that helps you observe patterns in how the model produces its outputs

Open the black box; explore internal representations

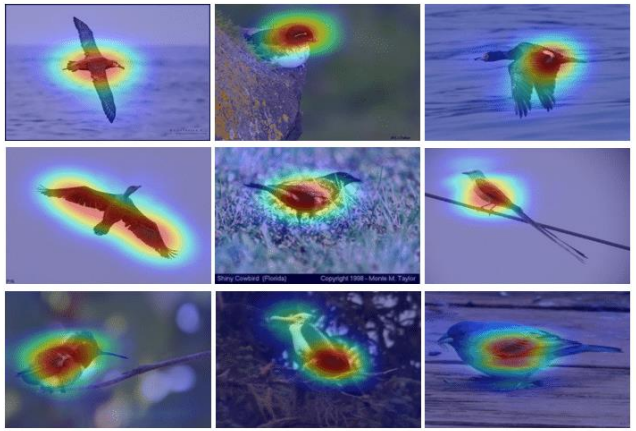
Assignment > Phase II What did the model learn?

Design a **visualization** that **helps you observe patterns** in how the model produces its outputs.

What should the visualization look like?

Wrong

Here are some saliency maps



Correct

Use visualization to explain behaviors

Example:

- Phase I: The model makes more mistakes on certain backgrounds.
- Phase II: Does the model rely on background features when producing these predictions?

More examples in the upcoming sessions

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AI4VIS

AI4VIS

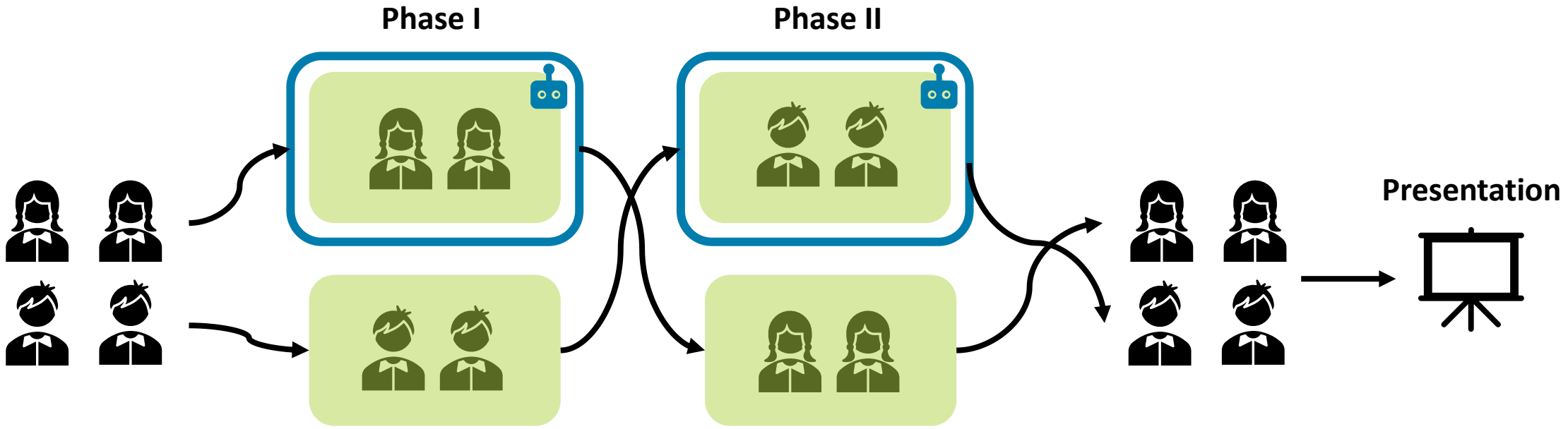
Warmup Discussion

- Do you collaborate with LLMs to perform tasks?
- Have you ever created charts to explore or communicate your data?
- Have you tried to leverage generative AI to assist you in designing visualizations?
  - Brainstorming and sketching designs
  - Provide your data => generate a visualization
  - Provide/describe your data/design => generate a visualization code
  - Annotate/explain your visualization
  - Transform your data
  - …

AI can support **VIS** in such aspects!

Assignment > AI4VIS

Explore how Generative AI can support and influence the visualization design and analysis process.



Assignment > AI4VIS

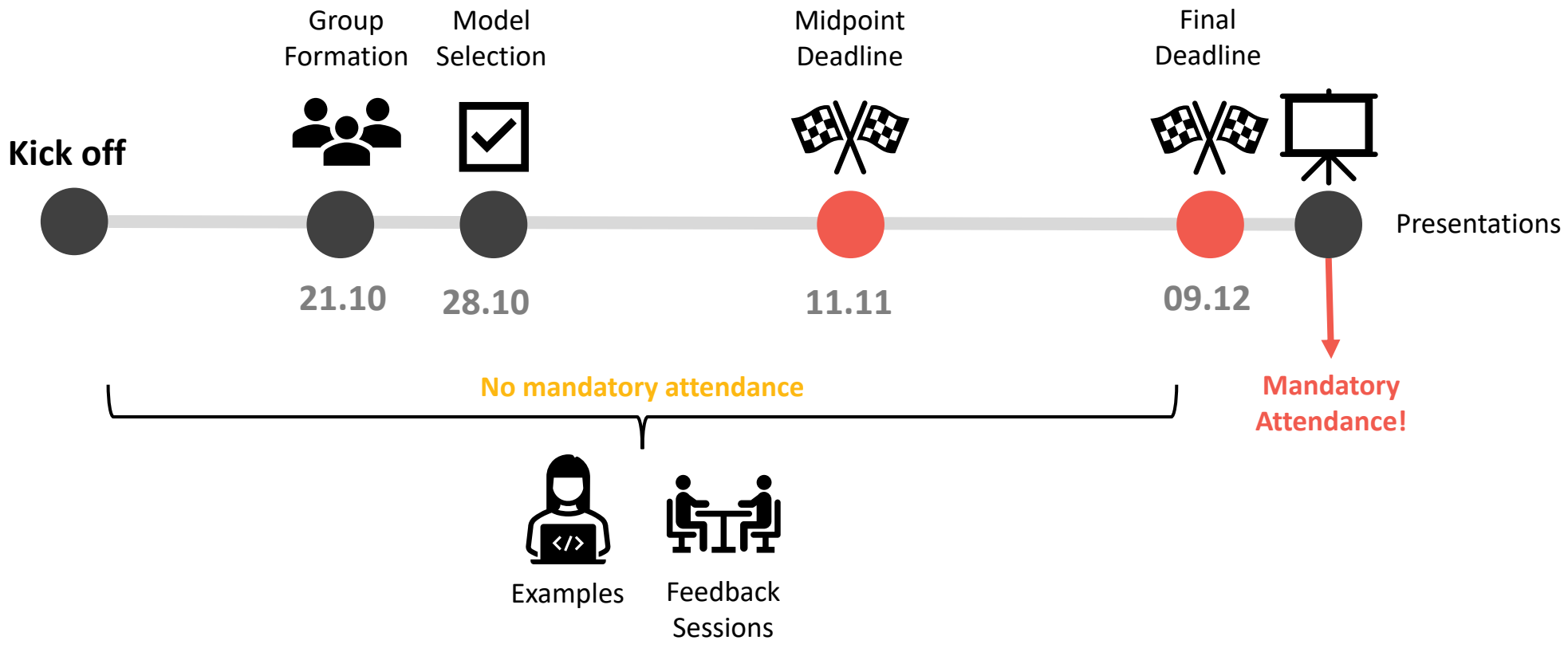
Explore how Generative AI can support and influence the visualization design and analysis process.

- **Record**:
  - The important prompts used
  - Which suggestions were accepted, rejected, or refined, and why
- **Report**:
  - Failure patterns observed during AI-assisted design & analysis
  - Emerging best practices for using AI effectively during visualization and analysis

Timeline

No deadline extensions!! Our timeline is tight

Keep up with recorded labs and lectures to complete the assignment successfully



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JKU Visual Data Science Lab

Questions